# $\mathrm{ModuleSVP}_{\infty}^{(2)}(\mathbb{Z}[\zeta_{2^k}])$ Is NP-Hard

**Abstract**

We prove deterministic NP-completeness of exact shortest-vector decision in the coefficient maximum norm for full-rank free rank-two modules over power-of-two cyclotomic rings. Our reduction from binary Subset Sum uses an explicit multiplication operator to force the ring multiplier of every short vector to be a signed monomial. The construction has polynomial binary size and also yields NP-hardness of exact shortest-vector search.

## 1 Introduction

Lattice-based cryptography is a central approach to post-quantum security, as illustrated by the standardization of ML-KEM and ML-DSA [10, 11]. Its security relies on the presumed hardness of lattice problems and related average-case problems, with worst-case-to-average-case reductions providing connections between them. Regev's quantum reduction from worst-case lattice approximation problems to learning with errors is a fundamental example [12]. The complexity of basic lattice problems depends on the norm and the required approximation factor. Van Emde Boas established deterministic NP-hardness for exact closest-vector problems and exact maximum-norm SVP [13]. For Euclidean SVP, Ajtai proved NP-hardness under randomized reductions [1], and Micciancio proved randomized hardness of approximation within every fixed factor strictly between one and $\sqrt{2}$ [9]. Recent preprints of Wan report deterministic hardness for exact Euclidean SVP and for approximation within every fixed constant factor greater than one [14, 15].

Algebraic structure permits compact representations and efficient arithmetic in lattice-based cryptography. It also imposes a substantive restriction on lattice hardness: a module contains all ring multiples of each vector, so a reduction must control more vectors than the integer combinations used in an ordinary lattice encoding. The ring and module worst-case-to-average-case reductions [7, 5] motivate studying these structured families. General lattice NP-hardness does not automatically persist under their algebraic restrictions.

Several recent results address this question for structured lattices. Liu, Feng, and Pan prove deterministic NP-completeness for exact SVP in the canonical Euclidean norm on free rank-two modules over $\mathbb{Z}[\zeta_q]$, where $q \equiv 3 \pmod 4$ is prime [6]. Martin gives a dimension-preserving reduction from general Euclidean lattice problems to ideal-lattice problems in varying totally real monogenic orders [8]. Wan proves deterministic NP-completeness for coefficient Euclidean SVP on cyclic lattices associated with $\mathbb{Z}[X]/(X^N - 1)$, where $N = q - 1$ for varying odd primes [16]. These results concern different algebraic families or norms from the one studied here.

We prove hardness for free rank-two modules over the prescribed power-of-two cyclotomic rings

$$R_n = \mathbb{Z}[X]/(X^n + 1) \cong \mathbb{Z}[\zeta_{2^k}], \qquad n = 2^{k-1}, \quad k \geq 1.$$

For $f = \sum_{i=0}^{n-1} f_i X^i$, define the coefficient maximum norm and its extension to two coordinates by

$$\|f\|_\infty = \max_{0 \leq i < n} |f_i|, \qquad \|(f, g)\|_\infty = \max\{\|f\|_\infty, \|g\|_\infty\}.$$

If $G \in R_n^{2\times 2}$ has nonzero determinant, put

$$\Lambda(G) = GR_n^2, \qquad \lambda_{1,\infty}(\Lambda(G)) = \min_{0 \neq v \in \Lambda(G)} \|v\|_\infty .$$

**Theorem 1.1.** *Given a power of two $n$, a matrix $G \in R_n^{2\times 2}$ with $\det G \neq 0$, and an integer $T \geq 0$, deciding whether $\lambda_{1,\infty}(\Lambda(G)) \leq T$ is NP-complete under deterministic polynomial-time many-one reductions. Finding an exact shortest nonzero vector in this family is NP-hard under deterministic polynomial-time Turing reductions.*

The module rank is fixed at two, while the ring degree and the decision radius vary with the input. All polynomial coefficients are explicitly represented binary integers. For a Subset Sum instance with $s$ weights, our reduction produces $n = O(s)$ and has polynomial binary size. Its YES instances have first minimum exactly $M$, whereas its NO instances have first minimum at least $M + 1$. Thus the theorem establishes exact hardness in the coefficient maximum norm; this additive separation does not give a fixed relative approximation factor.

To explain the proof, first recall the Boolean encoding available for ordinary lattices. Let $(w_1, \ldots, w_s; B)$ be a Subset Sum instance with positive integer weights and nonnegative target, and set

$$W = \sum_{j=1}^{s} w_j, \qquad C = 2(W + B + 1), \qquad \mathbf{1} = (1, \ldots, 1) \in \mathbb{Z}^s.$$

Consider the full-rank integer lattice

$$\mathcal{L}_{\mathrm{cl}} = \left\{ \left( 2a - b\mathbf{1},\ b,\ 2\Big(\sum_{j=1}^{s} w_j a_j - Bb\Big) + Cc \right) : a \in \mathbb{Z}^s,\ b, c \in \mathbb{Z} \right\} \subseteq \mathbb{Z}^{s+2}. \tag{1.1}$$

Its basis has determinant of absolute value $2^s C$ and polynomial binary length. A vector of norm at most one has $b \in \{0, \pm 1\}$. If $b = 0$, its first block forces $a = 0$ and its last coordinate forces $c = 0$. A nonzero short vector can therefore be negated to have $b = 1$. Its first block then gives $a \in \{0, 1\}^s$. The last coordinate is even and hence zero. Since

$$\left| 2\left( \sum_j w_j a_j - B \right) \right| \leq 2(W + B) < C,$$

this equality forces $c = 0$ and $\sum_j w_j a_j = B$. Conversely, every satisfying assignment gives a vector of norm one. This embedding gives a direct deterministic reduction from binary Subset Sum to maximum-norm SVP.

Our module construction retains the Boolean expression as $2a - ub$, where $a, b \in R_n$ and $u$ marks the positions of the weights. The additional difficulty is that the homogenizing multiplier $b$ is now a polynomial. Even a bound $\|b\|_\infty \leq 1$ allows many coefficients to be nonzero. We construct a multiplier filter which instead forces every nonzero short multiplier to belong to

$$\mathcal{U}_n = \{\varepsilon X^j : \varepsilon \in \{-1, 1\},\ 0 \leq j < n\}.$$

These units act by signed permutations of coefficients. Multiplying by $b^{-1}$ therefore preserves the norm and reduces the problem to the Boolean encoding with $b = 1$.

The filter uses $D = 1 + X^{n/2}$ to control the total absolute coefficient mass in each opposite pair. A second multiplication operator is skew-adjoint, meaning that its adjoint is its negative for the coefficient inner product. The short-vector inequalities make certain products nonpositive, while skew-adjointness makes their sum zero. Each product must consequently vanish. Powers of five separate the individual shift tests, and tests at both positions of an occupied pair force every other coefficient to vanish. After this normalization, reversed weights encode the subset sum in a constant coefficient, and an opposite coefficient imposes the reverse inequality. Together they force equality with the target.

Section 2 fixes the algebraic and computational definitions. Section 3 proves multiplier rigidity. Section 4 gives the reduction, proves its correctness and polynomial size, and derives witness recovery and search hardness. Section 5 concludes with the scope of the construction.

# 2 Preliminaries

## 2.1 Cyclotomic number fields and coefficient arithmetic

An algebraic number field is a finite extension of $\mathbb{Q}$. Its ring of integers consists of the elements satisfying a monic polynomial with integer coefficients. In this paper, for a power of two $n$, we use

$$\zeta_{2n} = e^{\pi i/n}, \qquad \mathbb{K}_n = \mathbb{Q}(\zeta_{2n}), \qquad R_n = \mathbb{Z}[\zeta_{2n}] \cong \mathbb{Z}[X]/(X^n + 1).$$

Here $R_n$ is the full ring of integers of $\mathbb{K}_n$ [17]. The polynomial $X^n + 1$ is irreducible: after the substitution $X \mapsto X + 1$, its constant coefficient is two, all intermediate coefficients are even, and its leading coefficient is one, so Eisenstein's criterion at two applies. Thus

$$[\mathbb{K}_n : \mathbb{Q}] = n, \qquad R_n = \bigoplus_{i=0}^{n-1} \mathbb{Z}X^i$$

and $R_n$ is an integral domain. We identify $X$ with $\zeta_{2n}$ under the displayed isomorphism. All ring products are reduced using $X^n = -1$.

For $f \in R_n$, write its unique coefficient representation as

$$f = \sum_{i=0}^{n-1} f_i X^i, \qquad \operatorname{coeff}_n(f) = (f_0, \ldots, f_{n-1})^T.$$

For two coordinates we concatenate the arrays:

$$\operatorname{coeff}_{2n}(f, g) = \begin{pmatrix} \operatorname{coeff}_n(f) \\ \operatorname{coeff}_n(g) \end{pmatrix}.$$

The coefficient norms and inner product, also defined for real arrays, are

$$\|f\|_1 = \sum_{i=0}^{n-1} |f_i|, \qquad \|f\|_\infty = \max_{0 \le i < n} |f_i|, \qquad \langle f, g \rangle = \sum_{i=0}^{n-1} f_i g_i. \tag{2.1}$$

The coefficient maximum norm on a pair is the maximum of the two coordinate maximum norms.

For any integer index $i$, use Euclidean division $i = qn + r$, where $q \in \mathbb{Z}$ and $0 \le r < n$, to define

$$\widetilde{f}_i = (-1)^q f_r, \qquad \widetilde{f}_{i+n} = -\widetilde{f}_i.$$

This antiperiodic extension records the signs in the shift identity

$$(X^d f)_i = \widetilde{f}_{i-d} \qquad (d \in \mathbb{Z}). \tag{2.2}$$

In particular, multiplication by a signed monomial is a signed permutation of the coefficient array. The set

$$\mathcal{U}_n = \{\varepsilon X^j : \varepsilon \in \{-1, 1\},\ 0 \le j < n\} \tag{2.3}$$

therefore consists of norm-preserving units. Their inverses are integral ring elements because $X^{-j} = -X^{n-j}$ for $1 \le j < n$.

The involution induced by complex conjugation is

$$f^* = f(X^{-1}), \qquad X^{-1} = -X^{n-1}.$$

It is also the adjoint of multiplication for the coefficient inner product:

$$\langle fg, h\rangle = \langle g, f^* h\rangle, \qquad \|fg\|_\infty \le \|f\|_1 \, \|g\|_\infty \,. \tag{2.4}$$

Indeed, the transpose of the signed shift $g \mapsto X^i g$ is $h \mapsto X^{-i} h$, so

$$\langle fg, h\rangle = \sum_i f_i \langle X^i g, h\rangle = \sum_i f_i \langle g, X^{-i} h\rangle = \langle g, f^* h\rangle.$$

The second identity follows from the triangle inequality:

$$\|fg\|_\infty = \left\| \sum_i f_i X^i g \right\|_\infty \le \sum_i |f_i| \left\| X^i g \right\|_\infty = \|f\|_1 \, \|g\|_\infty \,.$$

For a linear operator $A$ on the real coefficient space, its adjoint $A^\dagger$ is defined by

$$\langle Ax, y\rangle = \langle x, A^\dagger y\rangle.$$

The operator is called *skew-adjoint* if

$$A^\dagger = -A.$$

Thus multiplication by a polynomial $K$ is skew-adjoint precisely when $K^* = -K$. In that case, symmetry of the real inner product gives

$$\langle b, Kb\rangle = \langle Kb, b\rangle = \langle b, K^* b\rangle = -\langle b, Kb\rangle, \qquad \text{hence} \qquad \langle b, Kb\rangle = 0. \tag{2.5}$$

These adjoints concern the coefficient inner product used throughout the paper.

## 2.2 Lattices, SVP, and rank-two modules

A lattice with integer basis matrix $B \in \mathbb{Z}^{N \times d}$, whose columns are linearly independent, is

$$\Lambda(B) = \{Bz : z \in \mathbb{Z}^d\} \subseteq \mathbb{R}^N.$$

Its rank is $d$, and it is full rank in its ambient space when $d = N$. For the maximum norm, the first minimum is

$$\lambda_{1,\infty}(\Lambda(B)) = \min_{0 \neq v \in \Lambda(B)} \max_{1 \le i \le N} |v_i|.$$

The decision version of exact $\ell_\infty$-SVP asks, given $(B, T)$ with integer $T \ge 0$, whether $\lambda_{1,\infty}(\Lambda(B)) \le T$. The search version asks for a nonzero lattice vector attaining this minimum.

In the structured problem studied here, a module basis is a matrix

$$G = (g_1 \; g_2) \in R_n^{2\times 2}, \qquad \det G \neq 0,$$

and its module is

$$\Lambda(G) = R_n g_1 + R_n g_2 = \left\{ G \begin{pmatrix} a \\ b \end{pmatrix} : a, b \in R_n \right\}. \tag{2.6}$$

Since $G$ is invertible over $\mathbb{K}_n$, its two columns are $R_n$-linearly independent. Every element has unique multipliers $(a, b)$, so the module is free of rank two. The vectors $X^j g_i$, for $i \in \{1, 2\}$ and $0 \le j < n$, form an integer basis of its coefficient lattice in $\mathbb{Z}^{2n}$. They span because $1, X, \ldots, X^{n-1}$ span $R_n$; a linear relation among them would give $G(a, b)^T = 0$, forcing $a = b = 0$.

For explicit computations, let $C_n(f)$ be the integer matrix of multiplication by $f$:

$$(C_n(f))_{ij} = \begin{cases} f_{i-j}, & i \geq j, \\ -f_{n+i-j}, & i < j, \end{cases} \qquad 0 \leq i, j < n.$$

Its $j$th column is $\mathrm{coeff}_n(X^j f)$. The coefficient basis matrix and its defining identity are

$$\widehat{G} = \begin{pmatrix} C_n(G_{11}) & C_n(G_{12}) \\ C_n(G_{21}) & C_n(G_{22}) \end{pmatrix}, \qquad \mathrm{coeff}_{2n}\left(G\begin{pmatrix} a \\ b \end{pmatrix}\right) = \widehat{G}\begin{pmatrix} \mathrm{coeff}_n(a) \\ \mathrm{coeff}_n(b) \end{pmatrix}. \tag{2.7}$$

In particular, $\widehat{G}$ is a nonsingular $2n \times 2n$ integer matrix. Conversely, singularity of $G$ over $\mathbb{K}_n$ gives a nonzero kernel vector with rational coefficient arrays and hence a rational kernel vector for $\widehat{G}$. Thus the nonsingularity conditions are equivalent and can be checked by exact rational linear algebra.

**Definition 2.1** (Exact rank-two module SVP)**.** The input to $\mathrm{ModuleSVP}_\infty^{(2)}$ is

$$n \text{ a power of two}, \qquad G \in R_n^{2\times 2},\ \det G \neq 0, \qquad T \in \mathbb{Z}_{\geq 0}.$$

The entries of $G$ are explicit length-$n$ coefficient arrays, and all integers are encoded in binary. The question is whether

$$\exists\, v \in \Lambda(G) \setminus \{0\} : \|v\|_\infty \leq T,$$

or equivalently $\lambda_{1,\infty}(\Lambda(G)) \leq T$. Malformed inputs are rejected. The corresponding exact search problem takes $(n, G)$ and asks for a shortest nonzero module vector.

**Proposition 2.2.** *The language* $\mathrm{ModuleSVP}_\infty^{(2)}$ *belongs to* NP.

*Proof.* Let $N = 2n$ and let $A \geq 1$ bound the absolute entries of $\widehat{G}$. A certificate consists of integer vectors $v, t$ satisfying

$$0 < \|v\|_\infty \leq T, \qquad \widehat{G}t = v.$$

We verify these conditions by integer arithmetic. It remains to show that every YES instance has such a certificate of polynomial length. Choose any nonzero module vector $v$ of norm at most $T$. Its unique coefficient multiplier vector is an integer $t$. By Cramer's rule, for each coordinate,

$$t_i = \frac{\det \widehat{G}_i(v)}{\det \widehat{G}},$$

where $\widehat{G}_i(v)$ replaces column $i$ by $v$. Put $U = \max\{A, T, 1\}$. Every column of $\widehat{G}_i(v)$ has Euclidean norm at most $\sqrt{N}\,U$, so Hadamard's inequality gives

$$|\det \widehat{G}_i(v)| \leq N^{N/2} U^N.$$

The nonzero denominator is an integer of absolute value at least one. Thus each $t_i$ has $O(N(\log(N+1) + \log(U+1)))$ bits. The vector $v$ has $O(N \log(T+2))$ bits. Both lengths are polynomial in the explicit input length, and checking the displayed equations is polynomial time. Input validity can also be checked in polynomial time, using the nonsingularity test above. If $T = 0$, there is no nonzero witness. □

### 2.3 Membership and recovery of multipliers

**Lemma 2.3.** *Given valid $(n, G)$ and $v \in R_n^2$, one can decide in polynomial bit time whether $v \in \Lambda(G)$ and, if so, recover its unique multipliers $(a, b) \in R_n^2$.*

*Proof.* Use (2.7) to solve

$$\widehat{G}t = \mathrm{coeff}_{2n}(v)$$

over $\mathbb{Q}$. There is a unique rational solution because $\widehat{G}$ is nonsingular. It is integral if and only if $v$ belongs to the module: an integral solution gives the two coefficient arrays of $a, b$, while every module representation gives such an integral solution. Exact Gaussian elimination, or fraction-free elimination followed by divisibility tests, computes this solution in polynomial bit time [2]. The same determinant bounds used in Proposition 2.2, with $T$ replaced by the largest absolute coefficient of $v$, bound the binary lengths of the numerators and denominator. Splitting an integral $t$ into its two length-$n$ blocks recovers $a, b$. □

### 2.4 The subset-sum problem

A binary Subset Sum instance consists of

$$s \geq 1, \qquad (w_1, \ldots, w_s) \in \mathbb{Z}_{>0}^s, \qquad B \in \mathbb{Z}_{\geq 0}.$$

It is a YES instance precisely when

$$\sum_{j=1}^{s} w_j \xi_j = B \qquad \text{for some } \xi = (\xi_1, \ldots, \xi_s) \in \{0,1\}^s. \tag{2.8}$$

All weights and the target are encoded in binary. The empty subset is allowed, and repeated weights are distinguished by their positions. This problem is NP-complete under deterministic polynomial-time many-one reductions [4, 3]; its positive-target restriction already has this property. A proposed Boolean word is verified by binary addition in polynomial time.

For the size estimates, use

$$\mathrm{bits}(t) = \max\{1, \lceil \log_2(t+1) \rceil\} \quad (t \in \mathbb{Z}_{\geq 0}), \qquad \ell = s + \mathrm{bits}(B) + \sum_{j=1}^{s} \mathrm{bits}(w_j).$$

This is a binary input-length measure up to constant encoding overhead. Throughout the reduction, polynomial time refers to bit operations in $\ell$, rather than to the numerical values of the weights.

## 3 Rigidity of short multipliers

Fix $n = 2m$ with $n$ a power of two and $m \geq 2$. Define

$$D = 1 + X^m, \qquad S^{(d)} = X^d - X^{-d} \quad (1 \leq d < m), \qquad K = \sum_{d=1}^{m-1} 5^{d-1} S^{(d)}. \tag{3.1}$$

For $0 \leq j < m$, call $\{j, j+m\}$ an opposite pair. It is occupied by $b$ when $(b_j, b_{j+m}) \neq (0, 0)$.

**Lemma 3.1.** *For every $b \in R_n$,*

$$\|Db\|_\infty = \max_{0 \le j < m} (|b_j| + |b_{j+m}|) \ge \|b\|_\infty . \tag{3.2}$$

*Moreover,*

$$D^*D = 2, \qquad K^* = -K, \qquad \langle Db, DKb \rangle = 0, \tag{3.3}$$

$$K_0 = K_m = (DK)_0 = (DK)_m = 0, \qquad \|DK\|_1 \le 5^{m-1} - 1. \tag{3.4}$$

*Proof.* Since $X^{2m} = -1$, the coefficients of $X^m b$ at positions $j$ and $j+m$ are $-b_{j+m}$ and $b_j$, respectively. Hence multiplication by $D$ acts on each opposite pair as

$$((Db)_j, (Db)_{j+m}) = (b_j - b_{j+m}, b_j + b_{j+m}).$$

For real $c, e$,

$$\begin{aligned} \max\{|c-e|, |c+e|\}^2 &= \max\{c^2 + e^2 - 2ce, c^2 + e^2 + 2ce\} \\ &= c^2 + e^2 + 2|ce| = (|c| + |e|)^2. \end{aligned}$$

Both sides before squaring are nonnegative, so taking square roots and then the maximum over all opposite pairs proves the equality in (3.2). Each coefficient of $b$ belongs to one pair and is bounded by that pair's absolute sum, proving the final inequality.

The relation $X^{2m} = -1$ also gives $X^{-m} = -X^m$. Therefore

$$D^* = 1 - X^m, \qquad D^*D = (1 - X^m)(1 + X^m) = 1 - X^{2m} = 2.$$

For every $1 \le d < m$, applying the involution gives

$$(S^{(d)})^* = X^{-d} - X^d = -S^{(d)},$$

so $K^* = -K$ by linearity. Symmetry of the coefficient inner product and (2.4) imply

$$\langle b, Kb \rangle = \langle Kb, b \rangle = \langle b, K^*b \rangle = -\langle b, Kb \rangle,$$

whence $\langle b, Kb \rangle = 0$. Using $D^*D = 2$ now yields

$$\langle Db, DKb \rangle = \langle b, D^*DKb \rangle = 2\langle b, Kb \rangle = 0.$$

For $1 \le d < m$, we have $X^{-d} = -X^{n-d}$ and thus $S^{(d)} = X^d + X^{n-d}$. The two exponent ranges $1, \dots, m-1$ and $m+1, \dots, n-1$ are disjoint and avoid $0, m$. Consequently $K_0 = K_m = 0$. Applying the pair formula to $K$ gives

$$((DK)_0, (DK)_m) = (K_0 - K_m, K_0 + K_m) = (0, 0).$$

There are exactly two coefficients $5^{d-1}$ for each $d$, all positive. Therefore the geometric sum gives

$$\|K\|_1 = 2 \sum_{d=1}^{m-1} 5^{d-1} = 2\frac{5^{m-1} - 1}{5 - 1} = \frac{5^{m-1} - 1}{2}.$$

Finally, the triangle inequality and signed-shift invariance imply

$$\|DK\|_1 = \|K + X^m K\|_1 \le \|K\|_1 + \|X^m K\|_1 = 2\,\|K\|_1 = 5^{m-1} - 1.$$

This proves all the asserted identities and bounds. □

**Lemma 3.2** (Support rigidity)**.** *Suppose $b \in R_n$ satisfies $\|Db\|_\infty \le 1$ and $Kb$ vanishes at both positions of every occupied opposite pair of $b$. Then $b = 0$ or $b \in \mathcal{U}_n$.*

*Proof.* By (3.2), $|b_j| + |b_{j+m}| \le 1$ for every $0 \le j < m$. The two coefficients are integers, so each occupied pair contains exactly one coefficient equal to 1 or $-1$ and one zero coefficient. In particular, every coefficient of $b$, and hence every coefficient in its antiperiodic extension, belongs to $\{-1, 0, 1\}$. The shift formula in Section 2 therefore gives

$$(S^{(d)}b)_i = \widetilde{b}_{i-d} - \widetilde{b}_{i+d} \in \{-2, -1, 0, 1, 2\}.$$

Fix a position $i$ at which $Kb$ vanishes, and write $c_d = (S^{(d)}b)_i$. By (3.1),

$$0 = (Kb)_i = \sum_{d=1}^{m-1} 5^{d-1} c_d.$$

Reducing this equality modulo five shows that $c_1$ is divisible by five. As $-2 \le c_1 \le 2$, it follows that $c_1 = 0$. Removing that term and dividing by five gives the same statement for $c_2$; continuing in this way proves $c_d = 0$ for every $1 \le d < m$. Thus all individual shift tests vanish at every position where $Kb$ vanishes.

If $b \neq 0$, choose an occupied pair $\{j, j+m\}$ with $0 \le j < m$. By hypothesis, both positions satisfy every shift test. For $1 \le d < m$, the $d$th test at $j$ and the $(m-d)$th test at $j+m$ give

$$\begin{aligned} 0 &= (S^{(d)}b)_j = \widetilde{b}_{j-d} - \widetilde{b}_{j+d}, \\ 0 &= (S^{(m-d)}b)_{j+m} = \widetilde{b}_{j+d} - \widetilde{b}_{j+2m-d} = \widetilde{b}_{j+d} + \widetilde{b}_{j-d}. \end{aligned}$$

The final equality uses antiperiodicity with period $n = 2m$. Adding and subtracting these two equations yields $\widetilde{b}_{j-d} = \widetilde{b}_{j+d} = 0$. Modulo $2m$, the offsets $d$ for $1 \le d < m$ give $1, \dots, m-1$, while the offsets $-d$ give $m+1, \dots, 2m-1$. These disjoint ranges exhaust every residue except $0, m$. Translating by $j$ shows that all coefficients of $b$ outside $\{j, j+m\}$ vanish; the signs introduced by the antiperiodic extension do not affect vanishing. The chosen pair is occupied and has absolute coefficient sum at most one, so it contains exactly one nonzero coefficient, equal to 1 or $-1$. Hence $b \in \mathcal{U}_n$. □

**Proposition 3.3** (Multiplier rigidity)**.** *Let $H, V \in R_n$, and put*

$$\begin{aligned} &F = \|H\|_1, && \nu = \|V\|_1, && \rho = \|DK\|_1, \\ &L = F + \nu + 1, && M = 2(L\rho + \nu) + F + 1, && Q = MD + LDK. \end{aligned} \tag{3.5}$$

*For all $b, z \in R_n$,*

$$\|z\|_\infty \le 1, \quad \|Hz - (Q+V)b\|_\infty \le M \quad \Longrightarrow \quad b \in \{0\} \cup \mathcal{U}_n. \tag{3.6}$$

*These parameters are computable deterministically in time polynomial in $n$ and the binary lengths of the coefficients of $H, V$.*

*Proof.* Assume the two inequalities in (3.6), and set $t = \|Db\|_\infty$. Since $Db$ has integer coefficients, $t$ is a nonnegative integer. Equation (3.2) gives $\|b\|_\infty \le t$. Applying (2.4) to each product gives

$$\begin{aligned} \|Hz\|_\infty &\le \|H\|_1 \, \|z\|_\infty \le F, \\ \|DKb\|_\infty &\le \|DK\|_1 \, \|b\|_\infty \le \rho t, \\ \|Vb\|_\infty &\le \|V\|_1 \, \|b\|_\infty \le \nu t. \end{aligned}$$

From $Q = MD + LDK$ and the triangle inequality, we obtain

$$\begin{aligned} Mt = \|MDb\|_\infty &\le \|Hz - (Q+V)b\|_\infty + \|Hz\|_\infty + L\,\|DKb\|_\infty + \|Vb\|_\infty \\ &\le M + F + (L\rho + \nu)t. \end{aligned}$$

Thus

$$(M - L\rho - \nu)t \le M + F.$$

The coefficient on the left is positive, since $M - L\rho - \nu = L\rho + \nu + F + 1 > 0$. If $t \ge 2$, it follows that

$$2(M - L\rho - \nu) \le (M - L\rho - \nu)t \le M + F,$$

and hence

$$M \le 2(L\rho + \nu) + F = M - 1,$$

a contradiction. Therefore $t \le 1$. In particular, every occupied pair of $b$ contains a single sign coefficient, and the pair formula shows that $r = Db$ has two coefficients in $\{-1, 1\}$ at that pair and zero coefficients at every unoccupied pair.

Write $v = DKb$ and $e = Vb - Hz$. The preceding product bounds and $t \le 1$ give

$$\|e\|_\infty \le \|Vb\|_\infty + \|Hz\|_\infty \le \nu t + F \le \nu + F.$$

The second premise in (3.6) is equivalent to $|Mr_i + Lv_i + e_i| \le M$ for every coefficient position $i$. At a position with $r_i \ne 0$, we have $r_i \in \{-1, 1\}$, so

$$M + Lr_iv_i + r_ie_i = r_i(Mr_i + Lv_i + e_i) \le |Mr_i + Lv_i + e_i| \le M.$$

Subtracting $M$ and rearranging yields

$$Lr_iv_i \le -r_ie_i \le |e_i| \le \nu + F = L - 1 < L.$$

Since $L > 0$, this gives $r_iv_i < 1$. Both $r_i$ and $v_i$ are integers, so $r_iv_i \le 0$. At positions with $r_i = 0$ the product is zero. On the other hand, (3.3) says

$$\sum_{i=0}^{n-1} r_iv_i = \langle Db, DKb\rangle = 0.$$

Every summand is nonpositive, so each must be zero. Consequently $v_i = 0$ whenever $r_i \ne 0$.

For an occupied pair $\{j, j+m\}$, both $r_j$ and $r_{j+m}$ are nonzero, and therefore $v_j = v_{j+m} = 0$. Put $w = Kb$, so $v = Dw$. The pair formula gives

$$v_j = w_j - w_{j+m}, \qquad v_{j+m} = w_j + w_{j+m}.$$

Solving these two equations explicitly gives

$$w_j = \frac{v_j + v_{j+m}}{2} = 0, \qquad w_{j+m} = \frac{v_{j+m} - v_j}{2} = 0.$$

Thus $Kb$ vanishes at both positions of every occupied pair. Together with $\|Db\|_\infty = t \le 1$, this verifies both hypotheses of Lemma 3.2 and proves $b \in \{0\} \cup \mathcal{U}_n$.

For the complexity claim, suppose the coefficients of $H, V$ have at most $\tau \ge 1$ bits each. Their absolute sums $F, \nu$ are at most $n2^\tau$, so they have $O(\tau + \log n)$ bits. The largest coefficient of $K$ is $5^{m-2}$, whose binary length is $O(m) = O(n)$, and (3.4) gives $\rho \le 5^{m-1} - 1$. Thus $L$ has $O(\tau + \log n)$ bits and $M = 2L\rho + 2\nu + F + 1$ has $O(n + \tau + \log n)$ bits. Since

$$\|Q\|_\infty \le M\,\|D\|_\infty + L\,\|DK\|_\infty \le M + L\rho,$$

the coefficients of $Q$ satisfy the same bit bound. The coefficients of $K$ can be generated by successive multiplication by five, $DK$ by one signed shift and addition, and $F, \nu, \rho$ by summing absolute coefficients. The remaining formulas use integer additions and multiplications on numbers of the stated lengths. There are polynomially many such operations, so the entire construction runs in deterministic polynomial bit time. □

# 4 The reduction and proof of hardness

Given a SUBSET SUM instance $(w_1, \ldots, w_s; B)$, choose

$$m = 2^{\lceil \log_2 \max\{2,s\} \rceil}, \qquad n = 2m, \qquad W = \sum_{j=1}^{s} w_j. \tag{4.1}$$

Thus $m$ is the least power of two that is at least $\max\{2, s\}$. For $s \geq 2$ we have $s \leq m < 2s$, and hence $2s \leq n < 4s$. For $s = 1$, the choice is $m = 2$, $n = 4$. In all cases, $s \leq m$, $m \geq 2$, and $2s \leq n \leq 4s$. Define the following elements of $R_n$:

$$\begin{aligned} u &= \sum_{j=1}^{s} X^{j-1}, & h &= \sum_{j=1}^{s} w_j X^{-(j-1)}, & E &= 2B - W, \\ H &= (1 - X^m)h = D^*h, & V &= E(1 - X^m) = ED^*. \end{aligned} \tag{4.2}$$

The negative powers are interpreted in $R_n$, where $X^n = -1$. Apply Proposition 3.3 to $H, V$, using $F = \|H\|_1$, $\nu = \|V\|_1$, and $\rho = \|DK\|_1$, to obtain $L, M, Q$. Form

$$G = \begin{pmatrix} 2H & -Hu - Q - V \\ 2M & -Mu \end{pmatrix}. \tag{4.3}$$

The output of the reduction is $(n, G, M)$.

## 4.1 Boolean encoding and the equality test

**Lemma 4.1.** *For every $a \in R_n$,*

$$\|2a - u\|_\infty \leq 1 \quad \Longleftrightarrow \quad a = a_\xi := \sum_{j=1}^{s} \xi_j X^{j-1} \text{ for a unique } \xi \in \{0,1\}^s. \tag{4.4}$$

*For such $\xi$, put $A = \sum_{j=1}^{s} w_j \xi_j$ and $z_\xi = 2a_\xi - u$. Then*

$$\left\| (Hz_\xi - Q - V,\ Mz_\xi) \right\|_\infty = M + 2|A - B|. \tag{4.5}$$

*Proof.* Write $a = \sum_{i=0}^{n-1} a_i X^i$. At positions $0 \leq i < s$, the inequality in (4.4) is $|2a_i - 1| \leq 1$. Since $a_i$ is integral, this is equivalent to $a_i \in \{0, 1\}$. At positions $s \leq i < n$, it is $|2a_i| \leq 1$, which is equivalent to $a_i = 0$. These conditions prove both directions of (4.4) and the uniqueness of $\xi$. They also show that $z_\xi$ has coefficients $2\xi_j - 1 \in \{-1, 1\}$ at positions $j - 1$, $1 \leq j \leq s$, and zero elsewhere. In particular, $\|z_\xi\|_\infty = 1$.

Consider the Laurent expansion

$$f := hz_\xi = \sum_{j,k=1}^{s} w_j (2\xi_k - 1) X^{k-j}.$$

Every exponent belongs to $[-s + 1, s - 1] \subseteq [-m + 1, m - 1]$. A negative exponent $-r$, with $1 \leq r < m$, reduces to $-X^{n-r}$; a nonnegative exponent in this interval is already between 0 and $m - 1$. Thus, after reduction to the coefficient basis of $R_n$, only the terms with $j = k$ contribute at position 0, and no term contributes at position $m$. Consequently

$$f_0 = \sum_{j=1}^{s} w_j (2\xi_j - 1) = 2A - W, \qquad f_m = 0.$$

Multiplication by $D^* = 1 - X^m$ sends an opposite pair $(f_i, f_{i+m})$ to $(f_i + f_{i+m}, f_{i+m} - f_i)$. Since $Hz_\xi = D^* f$, the pair at $0, m$ is therefore

$$(Hz_\xi)_0 = f_0 + f_m = 2A - W, \qquad (Hz_\xi)_m = f_m - f_0 = -2A + W.$$

By (3.4), the coefficients of $DK$ at $0, m$ vanish. The definition $Q = MD + LDK$ therefore gives $Q_0 = Q_m = M$, whereas $V = E(1 - X^m)$ gives $V_0 = E$ and $V_m = -E$. Using $E = 2B - W$, the two corresponding coefficients of the first coordinate are

$$\begin{aligned} (Hz_\xi - Q - V)_0 &= (2A - W) - M - E = -M + 2(A - B), \\ (Hz_\xi - Q - V)_m &= (-2A + W) - M + E = -M - 2(A - B). \end{aligned} \tag{4.6}$$

For $M \geq 0$ and a real number $\delta$, $\max\{|-M+\delta|, |-M-\delta|\} = M + |\delta|$. Indeed, if $\delta \geq 0$, the second term is $M + \delta$ and bounds the first; the case $\delta \leq 0$ interchanges the terms. Thus the maximum absolute value in (4.6) is $M + 2|A - B|$.

At every other position, the coefficients of $MD$ and $V$ are zero. The convolution inequality (2.4) and $\|z_\xi\|_\infty = 1$ yield

$$|(Hz_\xi - Q - V)_i| \leq \|Hz_\xi\|_\infty + L\,\|DK\|_\infty \leq F + L\rho < M \qquad (i \notin \{0, m\}).$$

The last inequality is strict because $M - (F + L\rho) = L\rho + 2\nu + 1 > 0$. Finally, $\|Mz_\xi\|_\infty = M$. The two coefficients in (4.6) consequently determine the norm of the whole vector, proving (4.5). □

## 4.2 Correctness and size of the reduction

**Proposition 4.2.** *The matrix* (4.3) *has nonzero determinant, and*

$$\begin{aligned} \lambda_{1,\infty}(\Lambda(G)) &= M \quad \text{(YES)}, \\ M + 1 \leq \lambda_{1,\infty}(\Lambda(G)) &\leq 2M \quad \text{(NO)}. \end{aligned} \tag{4.7}$$

*The construction is deterministic polynomial time in the binary input length $\ell$, and its output length is $O(\ell^2)$.*

*Proof.* We first compute the norms used in the parameters and verify that $G$ defines a full-rank free rank-two module. For $2 \leq j \leq s$, the relations $X^{-(j-1)} = -X^{n-j+1}$ and $X^{m-(j-1)} = X^{m-j+1}$ give

$$H = w_1 - w_1 X^m - \sum_{j=2}^{s} w_j (X^{m-j+1} + X^{n-j+1}).$$

The exponents in the first nonconstant sum range from $m - s + 1$ to $m - 1$, and those in the second range from $n - s + 1$ to $n - 1$. Because $s \leq m$, these ranges lie in $[1, m-1]$ and $[m+1, n-1]$, respectively. They are disjoint and avoid $0, m$; for $s = 1$ both sums are empty. Every weight therefore occurs in two distinct coefficients, and

$$F = 2W, \qquad \nu = 2|E|. \tag{4.8}$$

Expanding the determinant of (4.3) gives

$$\det G = (2H)(-Mu) - (-Hu - Q - V)(2M) = 2M(Q + V).$$

Its factor $Q + V$ has constant coefficient $M + E$, since $Q_0 = M$. From $M = 2(L\rho + \nu) + F + 1$ we have $M > \nu = 2|E|$, so $M + E > 0$. In particular, $Q + V$ is a nonzero element of $R_n$. As $M > 0$ and $R_n$ is a domain, $\det G \neq 0$.

For arbitrary multipliers $a, b \in R_n$, put $z = 2a - ub$. Direct multiplication of the two rows of $G$ yields

$$G\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 2Ha - (Hu + Q + V)b \\ 2Ma - Mub \end{pmatrix} = \begin{pmatrix} Hz - (Q + V)b \\ Mz \end{pmatrix}. \tag{4.9}$$

Suppose this vector is nonzero and has norm at most $M$. Since $M > 0$, its second coordinate implies $\|z\|_\infty \le 1$. The first coordinate then supplies the other hypothesis of Proposition 3.3, so $b = 0$ or $b \in \mathcal{U}_n$. If $b = 0$, we have $z = 2a$. Every coefficient of $2a$ is an even integer with absolute value at most one, whence $a = z = 0$ and the vector in (4.9) is zero. This contradiction excludes $b = 0$.

It follows that $b = \varepsilon X^j \in \mathcal{U}_n$. Multiplication of both coordinates by $b^{-1}$ preserves the coefficient maximum norm and commutes with multiplication by $G$. With $a' = ab^{-1}$ we obtain

$$b^{-1}G\begin{pmatrix} a \\ b \end{pmatrix} = G\begin{pmatrix} a' \\ 1 \end{pmatrix} = \begin{pmatrix} H(2a' - u) - Q - V \\ M(2a' - u) \end{pmatrix}.$$

Its second coordinate gives $\|2a' - u\|_\infty \le 1$. Lemma 4.1 consequently identifies $a' = a_\xi$ for a unique $\xi \in \{0,1\}^s$ and gives

$$\left\| G(a', 1)^T \right\|_\infty = M + 2\left| \sum_{j=1}^{s} w_j \xi_j - B \right|.$$

The assumed bound by $M$ forces the absolute value to vanish. Thus every nonzero vector of norm at most $M$ determines a satisfying assignment and has norm exactly $M$. In particular, there is no nonzero vector of norm less than $M$ in any output lattice.

Conversely, if $\xi$ satisfies $\sum_j w_j \xi_j = B$, then Lemma 4.1 gives a lattice vector $G(a_\xi, 1)^T$ of norm $M$. It is nonzero because its second coordinate has norm $\|Mz_\xi\|_\infty = M > 0$. Hence a YES instance has first minimum exactly $M$. A NO instance has no nonzero vector of norm at most $M$; since all coefficients are integral, its first minimum is at least $M + 1$. The first column $(2H, 2M)^T$ is a nonzero lattice vector of norm $2M$: indeed, $\|H\|_\infty \le F < M$, whereas its second coordinate is the constant $2M$. This proves both bounds in (4.7).

To estimate the binary length, put

$$A_0 = W + B + 1, \qquad \beta = \lceil \log_2(W + B + 2) \rceil, \qquad q = 5^{m-1}.$$

Equations (3.4) and (4.8) imply

$$F + 1 = 2W + 1 < 2A_0, \qquad \nu = 2|2B - W| \le 4B + 2W < 4A_0, \qquad 0 \le \rho < q.$$

Moreover,

$$L = 2W + 2|2B - W| + 1 \le 4W + 4B + 1 < 4A_0.$$

Substitution in the definition of $M$ now gives, since $q \ge 5$,

$$\begin{aligned} M &= 2L\rho + 2\nu + F + 1 \\ &< 8A_0 q + 10A_0 \le 10A_0 q < 16A_0\, 5^{m-1}. \end{aligned}$$

Since $A_0 < 2^\beta$, this gives

$$\log_2 M < 4 + \beta + (m - 1)\log_2 5.$$

Thus $M$ has $O(m + \beta) = O(s + \beta)$ bits. The coefficients of the upper-right entry of $G$ satisfy

$$\begin{aligned} \|Hu + Q + V\|_\infty &\le \|H\|_1 \|u\|_\infty + M \|D\|_\infty + L \|DK\|_\infty + \|V\|_\infty \\ &\le F + M + L\rho + \nu < 2M, \end{aligned}$$

because $\|u\|_\infty = \|D\|_\infty = 1$ and $M - (F + L\rho + \nu) = L\rho + \nu + 1 > 0$. The coefficients of $2H$ are bounded by $2F < 2M$, those of $-Mu$ by $M$, and the remaining entry is the constant $2M$. Thus every basis coefficient is bounded in absolute value by $2M$. Writing all four entries in the coefficient basis requires $O(n \log(M + 2)) = O(s(s + \beta))$ bits, including the radius and the degree within the same bound.

For completeness, $s = O(\ell)$ because the input lists $s$ weights, and $\beta = O(\ell)$ because the binary lengths of $W$ and $B$ are bounded by the input length and the logarithm of the number of weights. Hence the output length is $O(\ell^2)$. The degree in (4.1) can be found by successive doubling, and the coefficients of $K$ by successive multiplication by five. Constructing $H, V, L, M, Q, G$ then uses polynomially many additions and multiplications of integers of polynomial binary length; polynomial products may be computed by ordinary coefficient convolution and reduction modulo $X^n + 1$. The entire construction is therefore deterministic polynomial time. □

*Proof of Theorem 1.1 (decision).* Proposition 4.2 gives a deterministic polynomial-time many-one reduction from binary Subset Sum. Malformed input encodings may be mapped to the fixed NO instance $(n, G, T) = (4, 2I_2, 1)$. Together with Proposition 2.2, this proves the asserted NP-completeness. □

### 4.3 Witness recovery and exact search

**Corollary 4.3.** *For a* Subset Sum *instance and its reduction output, the map*

$$(\xi, b) \longmapsto b\, G \begin{pmatrix} a_\xi \\ 1 \end{pmatrix}, \qquad \xi \in \{0,1\}^s, \quad \sum_{j=1}^{s} w_j \xi_j = B, \quad b \in \mathcal{U}_n, \tag{4.10}$$

*is a bijection onto the nonzero vectors in $\Lambda(G)$ of norm at most $M$. Its inverse is computable in polynomial bit time. In particular, if the instance has $N_{\mathrm{sol}}$ solutions, there are exactly $2nN_{\mathrm{sol}}$ such vectors.*

*Proof.* For every satisfying assignment $\xi$, the vector $G(a_\xi, 1)^T$ has norm $M$ by Lemma 4.1. Multiplication by $b \in \mathcal{U}_n$ preserves that norm and keeps the vector in $\Lambda(G)$, so (4.10) has the stated codomain. Conversely, let $v$ be any nonzero lattice vector of norm at most $M$. Since $\det G \neq 0$, it has unique multipliers $a, b \in R_n$ with $v = G(a, b)^T$. The soundness proof gives $b \in \mathcal{U}_n$ and $ab^{-1} = a_\xi$ for a satisfying assignment $\xi$. Therefore

$$v = G \begin{pmatrix} ba_\xi \\ b \end{pmatrix} = b\, G \begin{pmatrix} a_\xi \\ 1 \end{pmatrix},$$

proving surjectivity.

If the pairs $(\xi, b)$ and $(\xi', b')$ have the same image, then injectivity of the map defined by $G$ gives $(ba_\xi, b) = (b'a_{\xi'}, b')$. Equality of the second coordinates implies $b = b'$, and cancellation of this unit in the first gives $a_\xi = a_{\xi'}$, hence $\xi = \xi'$. The map is therefore injective. Since the $2n$ elements $\pm X^j$, $0 \le j < n$, are distinct in the coefficient basis, the counting assertion follows.

To compute the inverse, apply Lemma 2.3 to recover the unique multipliers $(a, b)$ from $v$. The coefficient array of $b$ identifies its sign and exponent, so multiplying $a$ by $b^{-1}$ is a signed shift. The first $s$ coefficients of $ab^{-1} = a_\xi$ are precisely $\xi_1, \ldots, \xi_s$. Exact linear algebra and the signed shift take polynomial bit time. Moreover, the input vector has $O(n \log(M + 2))$ bits, which is polynomial in $\ell$ by Proposition 4.2. Thus recovery is polynomial time in the original instance length as well. □

**Corollary 4.4.** *Exact shortest-vector search for the modules in Theorem 1.1 is NP-hard under deterministic polynomial-time Turing reductions.*

*Proof.* Given a Subset Sum instance, construct $(n, G, M)$ and query an exact shortest-vector oracle on $(n, G)$. Let $v$ be the returned shortest nonzero vector. The first column of $G$ has norm $2M$, so $\|v\|_\infty \leq 2M$ in every instance. Its $2n$ integer coefficients therefore require $O(n \log(M+2)) = O(\ell^2)$ bits. Reading the oracle output and computing its maximum coefficient norm are consequently polynomial-time operations. By (4.7), the original instance is YES exactly when $\|v\|_\infty \leq M$. This decides binary Subset Sum with one oracle call and polynomial additional computation. On YES instances, Corollary 4.3 also recovers a satisfying assignment from $v$ in polynomial bit time. □

# 5 Conclusion

We proved deterministic NP-completeness of exact SVP in the coefficient maximum norm for full-rank free rank-two modules over power-of-two cyclotomic rings. The reduction combines the Boolean encoding of Subset Sum with an explicit multiplier filter. This filter controls every ring multiplier of a short vector, permitting normalization by a signed monomial and recovery of a satisfying assignment. The same construction also establishes NP-hardness of exact shortest-vector search.

The proof uses individual coefficient bounds to turn skew-adjoint orthogonality into vanishing conditions. Extending this construction to finite-$p$ norms or to rank-one ideals would require a different way to obtain this rigidity or to enforce the Boolean constraints. A fixed relative approximation gap would likewise require stronger norm separation than the additive gap established here.

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